EXP NO 7️⃣
PROGRAM:-

```
clc; clear; close all;
N = 20;
b1 = firpm(N,[0 0.2 0.3 1],[1 1 0 0]);
b2 = fir2(N,[0 0.2 0.3 1],[1 1 0 0]);
freqz(b1,1,512,1000);
figure; freqz(b2,1,512,1000);
```

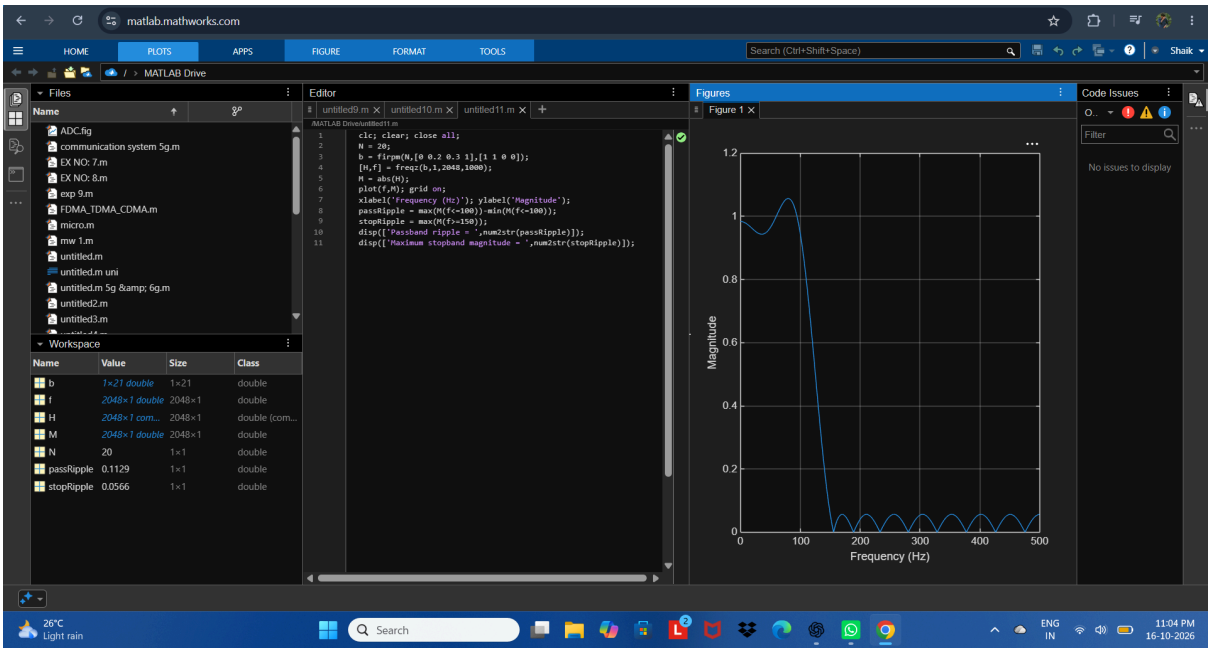
```
clc; clear; close all;
N = 20;
b1 = firpm(N,[0 0.2 0.3 1],[1 1 0 0]);
b2 = fir2(N,[0 0.2 0.3 1],[1 1 0 0]);
disp('Parks-McClellan coefficients:'); disp(b1);
disp('Frequency-sampling coefficients:'); disp(b2);
freqz(b1,1,512,1000);
```

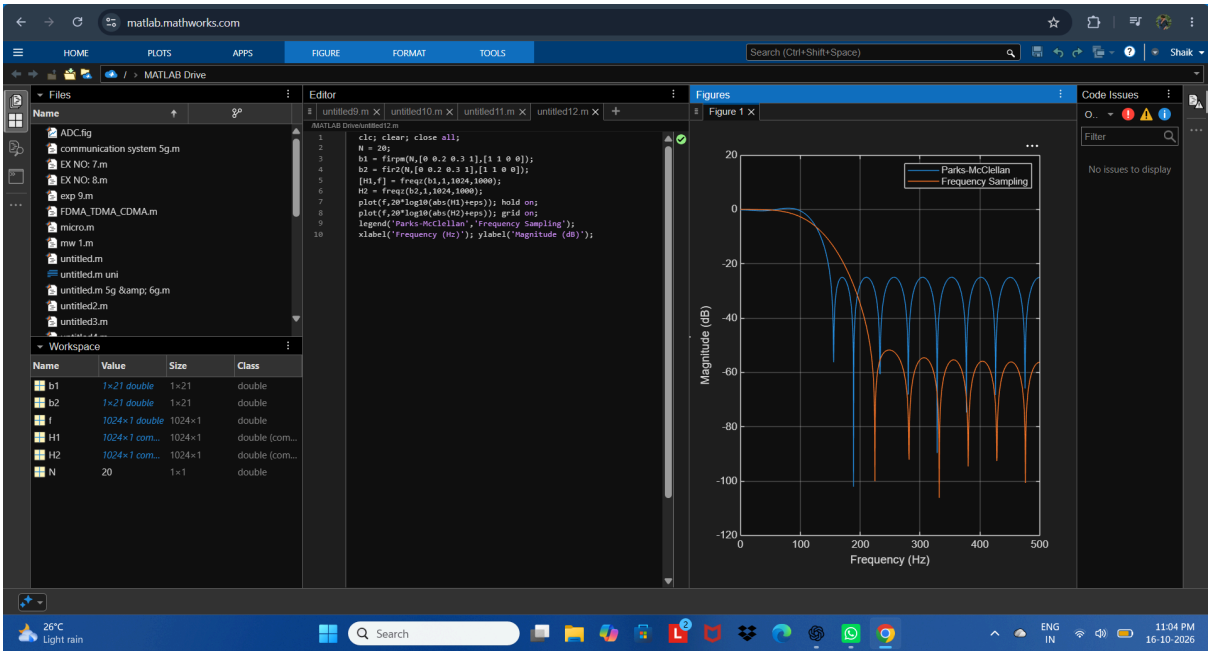
```
clc; clear; close all;
N = 20;
b = firpm(N,[0 0.2 0.3 1],[1 1 0 0]);
[H,f] = freqz(b,1,2048,1000);
M = abs(H);
plot(f,M); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude');
passRipple = max(M(f<=100))-min(M(f<=100));
stopRipple = max(M(f>=150));
disp(['Passband ripple = ',num2str(passRipple)]);
```

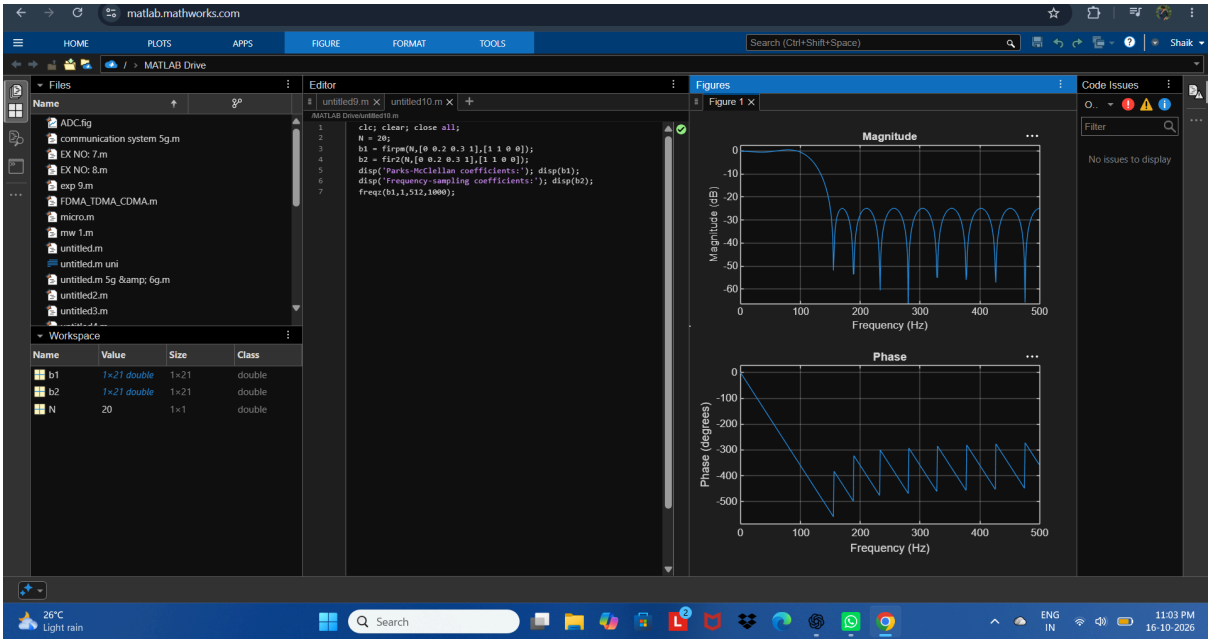
```
disp(['Maximum stopband magnitude =
',num2str(stopRipple)]);

clc; clear; close all;
N = 20;
b1 = firpm(N,[0 0.2 0.3 1],[1 1 0 0]);
b2 = fir2(N,[0 0.2 0.3 1],[1 1 0 0]);
[H1,f] = freqz(b1,1,1024,1000);
H2 = freqz(b2,1,1024,1000);
plot(f,20*log10(abs(H1)+eps)); hold on;
plot(f,20*log10(abs(H2)+eps)); grid on;
legend('Parks-McClellan','Frequency Sampling');
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
```


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b2 = fir2(N,[0 0.2 0.3 1],[1 1 0 0]);
[H1,f] = freqz(b1,1,1024,1000);
H2 = freqz(b2,1,1024,1000);
plot(f,20*log10(abs(H1)+eps)); hold on;
plot(f,20*log10(abs(H2)+eps)); grid on;
legend('Parks-McClellan','Frequency Sampling');
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
Parks-McClellan
Frequency Sampling
Magnitude (dB)
Frequency (Hz)
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b2 = fir2(N,[0 0.2 0.3 1],[1 1 0 0]);
disp('Parks-McClellan coefficients:'); disp(b1);
disp('Frequency-sampling coefficients:'); disp(b2);
freqz(b1,1,512,1000);
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